LGND

Where did this embedding come from?

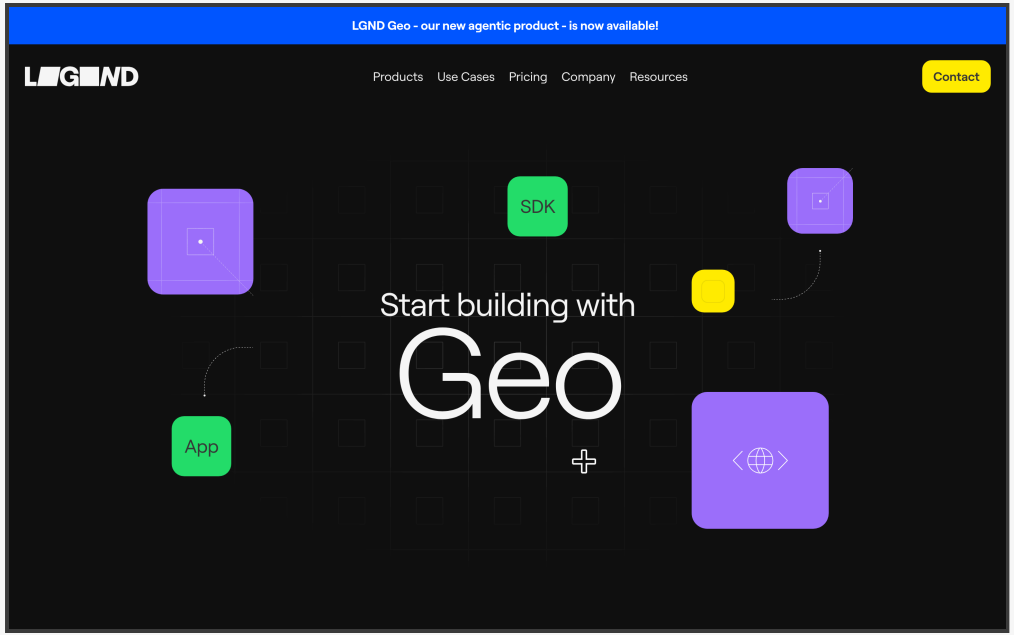
Provenance in the STAC embeddings extension

Matt Hanson

Geo-Embeddings Sprint II · Zürich · Oct 27, 2026

lgnd.ai · geoskeptic.dev

LGND Geo



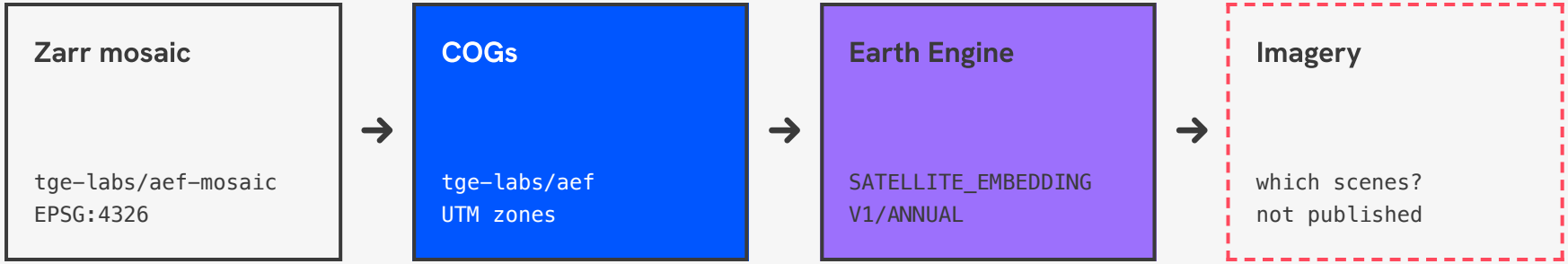
- embeddings as tools for your LLM
- MCP: Claude, ChatGPT, your agent
- Sentinel-2 · global · monthly
- NAIP · CONUS
- free

lgnd.ai/geo

An embedding is a **summary**.

A summary should point to its sources.

Follow AlphaEarth back



AlphaEarth can’t show its work

Would I explore with it?

Yes.

Would I use it as evidence?

No.

UNPUBLISHED INPUTS · CLOSED MODEL · CC-BY OUTPUTS

The file can describe itself. It can’t link.

Only the catalog can:

- `derived_from` the source Items
- link the model and version
- `processing:*`
- be **searched**

01

The extension today

`geo-embeddings/embeddings-stac-specification` · v0.0.1

What the draft describes

Product

`emb:type`	pixel or chip
`emb:dimensions`	vector length
`data_type`	numeric type
`gsd`	ground sampling distance
`emb:chip_layout`	chip size, stride, grid
`emb:temporal_resolution`	time represented
`emb:fit_for_use`	retrieval, classification…
`emb:uncertainty`	uncertainty information

Processing

`emb:runtime_parameters`	inference parameters
`emb:preprocessing`	pre-inference preparation
`emb:postprocessing`	output transformations
`emb:quantization`	vector encoding
`emb:energy`	compute and energy use
`processing:datetime`	production time
`processing:version`	software version

Links, and where fields go

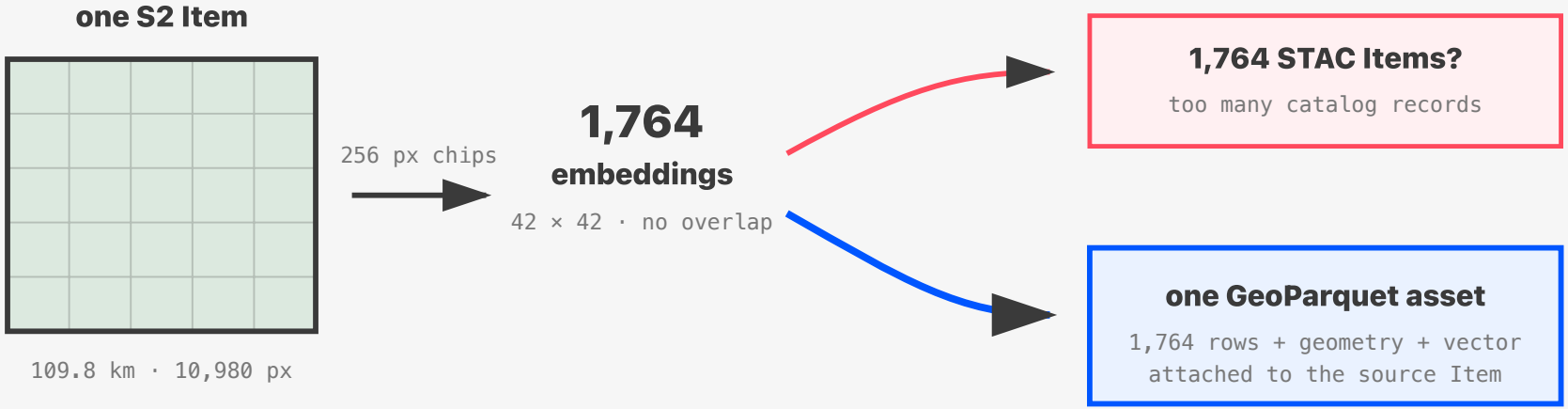
Link	Points to
`emb:model`	the model
`emb:source-data`	the source data
`emb:decoder`	a decoder
`emb:benchmark`	evaluation

Scope	`emb:*` allowed?
Collection	yes
Collection summaries	yes
Item properties	some
Item assets	**no**
`item_assets`	**no**

02

Two STAC patterns

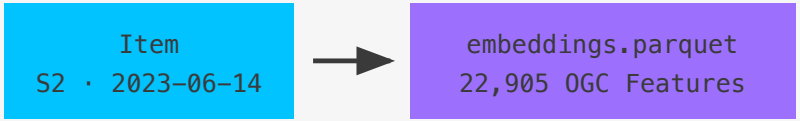
One scene is not 1,764 Items



Two products, two STAC patterns

One scene in, chips out

optical, per acquisition



an asset on the Item it came from — the provenance is the parent

---

Many scenes in, one surface out

per-pixel · annual · time series



Pattern 1: an asset on the S2 Item


```
{
  "type": "Feature",
  "stac_version": "1.0.0",
  "id": "S2B_53SKU_20240614_0_L2A",
  "collection": "sentinel-2-l2a",
  "properties": {
    "datetime": "2024-06-14T02:07:51.098Z",
    "platform": "sentinel-2b",
    "eo:cloud_cover": 7.65,
    …
  },
  "assets": {
    "red":    { "href": ".../B04.tif", … },
    "nir":    { "href": ".../B08.tif", … },
    "visual": { "href": ".../TCI.tif", … },
    …
```

```
    "embeddings-clay": {
      "href": ".../S2B_53SKU_20240614_0_L2A.parquet",
      "type": "application/vnd.apache.parquet",
      "roles": ["data", "embeddings"],
      "emb:type": "chip",
      "emb:dimensions": 256,
      "data_type": "float32",
      "emb:model": "clay-v1.5",
      "emb:normalization": "none",
      "emb:distance_metric": "cosine"
    }
  },
  "links": [
    { "rel": "derived_from",
      "href": ".../sentinel-2-l1c/items/S2B_53SKU_20240614_0_L1C" },
    …
  ]
}
```

Pattern 2: its own Item, linked back


```
{
  "type": "Feature",
  "id": "53SKU-2024-clay-composite",
  "properties": {
    "datetime": null,
    "start_datetime": "2024-03-16T00:00:00Z",
    "end_datetime": "2024-09-07T23:59:59Z",
    "emb:type": "chip",
    "emb:dimensions": 256
  },
  "assets": {
    "embeddings": {
      "href": ".../53SKU-2024.parquet",
      "type": "application/vnd.apache.parquet",
      "emb:model": "clay-v1.5",
      "emb:distance_metric": "cosine"
    }
  },
  "links": [
    { "rel": "derived_from",
      "href": ".../items/S2B_53SKU_20240316_0_L2A" },
    { "rel": "derived_from",
      "href": ".../items/S2B_53SKU_20240614_0_L2A" },
    { "rel": "derived_from",
      "href": ".../items/S2A_53SKU_20240907_0_L2A" },
    { "rel": "emb:model",
      "href": "https://huggingface.co/made-with-clay/Clay" },
    { "rel": "collection",
      "href": "./collection.json" }
  ]
}
```

03 What I'd change

1 · Allow `emb:*` on assets

```
STAC Item · one Sentinel-2 scene
├── visual                  image/tiff
├── embeddings-clay         application/vnd.apache.parquet
│   ├── emb:dimensions      1024
│   └── emb:model           clay-v1.5
└── embeddings-terramind    application/vnd.apache.parquet
    ├── emb:dimensions      768
    └── emb:model           terramind-v1
```

What goes where

Put on the Item or Collection	Put on the embedding asset
spatial and temporal extent	dimensions and data type
intended use	model identity and version
product-level provenance	normalization and distance metric
searchable product characteristics	chip layout and vector column
links to source Items	preprocessing and quantization

2 · Support both patterns

Pattern 1 · an asset on the source Item

Needs asset scope.

Pattern 2 · a derived embedding Item

Works today.

But `derived_from` doesn't scale

Instead:

- **snapshot** the source catalog: metadata, not data
- record **context geometry** + **time range**
- the sources become a query

3 · No sources? Say what you know.

- source datasets and date range
- model and processing version
- which inputs are missing or inaccessible
- licences and access restrictions
- a provenance status

04 Search

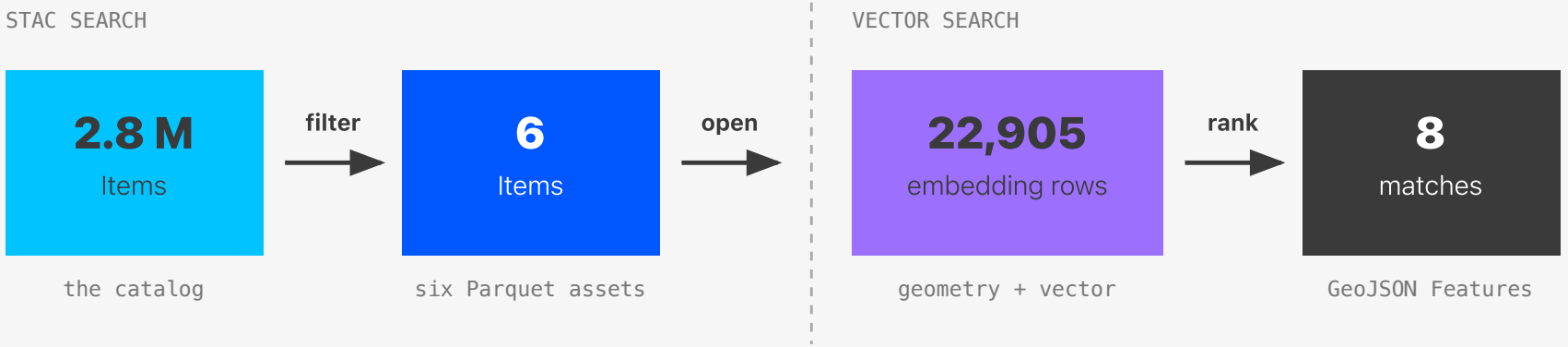
Search with CQL2

```
POST /embeddings/search
{
  "collections": ["sentinel-2-l2a"],
  "filter-lang": "cql2-json",
  "filter": { "op": "and", "args": [
    { "op": "s_intersects", "args": [{ "property": "geometry" }, { "bbox": [-119.0, 35.0, -117.5, 35.6] }] },
    { "op": "t_intersects", "args": [{ "property": "datetime" }, { "interval": ["2023-01-01", "2023-12-31"] }] },
    { "op": "<", "args": [{ "property": "eo:cloud_cover" }, 10] },
    { "op": ">", "args": [
      { "op": "cosine_similarity", "args": [{ "property": "embeddings-clay" }, [0.013, -0.225, 0.084, 0.019]] },
      0.8 ] }
  ]},
  "limit": 20
}
```

The response is a FeatureCollection

```
{
  "type": "FeatureCollection",
  "features": [{
    "type": "Feature",
    "id": "chip-903",
    "geometry": { "type": "Polygon", "coordinates": ["..."] },
    "properties": {
      "score": 0.87,
      "item": "S2B_11SMU_20230818",
      "asset": "embeddings-clay"
    },
    "links": [{ "rel": "item", "href": ".../items/S2B_11SMU_20230818" }]
  }]
}
```

Two passes: STAC, then DuckDB



1 · STAC search

place, time, cloud cover → Parquet hrefs

2 · DuckDB

`read_parquet(hrefs)` → cosine similarity → top *k*

Open questions

Too many sources	Is a snapshot Collection enough to reproduce the selection? Or does the scene filter (cloud cover, say) need recording too?
Which source, which row	`derived_from` is per Item. Pattern 1 rows can carry their `stac_item_id`. For S1 + S2 with different footprints in Pattern 2, I don't know what the convention should be.
The model changes	New asset, new Item, or new Collection? It's the embeddings version of calibration lineage in ARD.
Search	Would you use it? Or is STAC search then DuckDB enough? And in CQL2: what's the function called, how do you name an asset, and how do results get ranked?

Do not separate the summary from the evidence.



Thanks

geoskeptic.dev

SLIDES

geoskeptic.dev/talks

matt@lgnd.ai